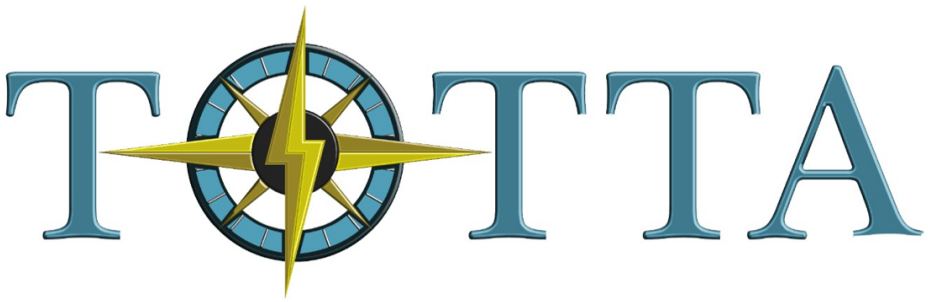
Stablecoins as Base Currency

For Sustainable and Inclusive Global Trade Finance

by

Subra Shankhar & Chris Sunderman





FOREWORD

by

Dr Tom James
CEO & CIO
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Global trade is entering one of the most significant periods of transformation since the introduction of containerisation and electronic banking. While goods continue to move across oceans much as they have for generations, the financial and digital infrastructure supporting those movements is being fundamentally reimagined.

For too long, international commerce has been constrained by fragmented payment systems, slow cross-border settlement, high transaction costs and limited access to finance for millions of small and medium-sized enterprises. At the same time, investors, policymakers and regulators increasingly expect capital to support not only economic growth, but also resilience, transparency and sustainable development.

Stablecoins represent an important opportunity within this transformation. Properly regulated and supported by robust governance, they have the potential to modernise the movement of value in much the same way that digital documentation is modernising the movement of information. Combined with advances in artificial intelligence, distributed ledger technology, digital identity and electronic transferable records, they can help create a more efficient, transparent and inclusive global trading system.

However, technology alone is not the solution. Sustainable progress will require trusted legal frameworks, sound regulation, strong risk management and close collaboration between governments, financial institutions, technology providers and the private sector. The objective should not simply be faster payments, but better markets—markets that are more resilient, more accessible and capable of directing capital towards productive economic activity.

In this discussion paper, Subra Shankhar and Chris Sunderman explore an important aspect of this evolving landscape by examining how regulated stablecoins may serve as an operational settlement layer for sustainable trade finance, particularly within emerging markets. Their work encourages policymakers, financial institutions and industry participants to consider how digital money can complement broader initiatives aimed at improving financial inclusion, reducing friction in cross-border commerce and strengthening global supply chains.



Whether every conclusion is ultimately realised is less important than ensuring that these conversations take place now. The future architecture of international trade is being designed today. Those who engage constructively in this dialogue will help shape a trading system that is more transparent, more inclusive and better equipped to meet the economic and sustainability challenges of the decades ahead.

I commend the authors for contributing to this important discussion and hope this paper stimulates further debate, research and practical collaboration across the global trade ecosystem.

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PREFACE

DIGITAL TRADE FINANCE

Global trade finance underpins economic stability, food security, and supply chain resilience, particularly in emerging markets where currency volatility, underdeveloped banking infrastructure, and high transaction costs constrain participation in global trade. Traditionally, trade finance has focused on facilitating trade flows and generating returns on equity, with limited emphasis on sustainability or financial inclusion.

As sustainability has become a core investment metric, with **ESG-aligned assets** projected to grow from approximately **US$18 trillion today to more than US$40 trillion by 2030**, investors are increasingly seeking opportunities that combine long-term financial returns with measurable environmental and social impact. Savvy and responsible investors now evaluate companies and platforms based on ESG transparency, digital maturity, and their capacity to integrate inclusive payment and settlement infrastructures that support SMEs and emerging market participants.

Trade finance is uniquely suited to stablecoin adoption because it is built around short-duration, self-liquidating transactions supported by extensive commercial documentation, making it one of the most natural environments for digital settlement using regulated stablecoins.

Beyond their technological innovation, regulated stablecoins address several long-standing inefficiencies in cross-border trade finance. By reducing settlement friction, releasing trapped liquidity, lowering idle capital requirements, and mitigating foreign exchange uncertainty, they improve capital efficiency across the trade ecosystem.

As digital assets such as stablecoins mature and various digital technologies such as AI and blockchain become mainstream, **trade finance is rapidly emerging as a strategic asset class for investors seeking sustainable and inclusive growth.**

This paper assumes a basic understanding of single-currency stablecoins. Basket-backed stablecoins will be the subject of a subsequent paper.

In this paper, we make the case that digital trade finance—powered by blockchain, AI, IoT, and in particular, regulated stablecoins—is not only transforming global commerce but also providing the monetary foundation for sustainable and inclusive investment strategies. This is not fundamentally a paper about stablecoins.



Accordingly, this paper is less about stablecoins as a technology than about improving capital efficiency in global trade finance. It argues that regulated stablecoins are the most practical operational settlement layer for achieving that objective, while complementing the broader evolution of digital trade ecosystems.

Chris and Subra

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Table of Contents

Foreword I

Preface III

1. DIGITAL TRADE FINANCE, STABLECOINS, AND SUSTAINABLE INVESTMENT 1

Promoting Transparency, Accountability, and Monetary Stability 1

Reducing Carbon Footprints and Operational Waste 1

Supporting Green Finance, Programmability, and Incentive Alignment 1

2. COMBATING GREENWASHING THROUGH AI, DATA ANALYTICS, AND TOKENIZED FLOWS 2

3. STABLECOINS, FOOD SECURITY, AND INVESTMENT RISK IN COMMODITY TRADE 2

4. STABLECOINS, SMES, AND EMERGING MARKETS: INCLUSION & SUSTAINABILITY 3

Lowering Barriers to Global Trade Participation 3

Financial Inclusion, ESG, and Digital Trade Ecosystems 3

Fintech Collaboration, CBDCs, and the Sustainable Investment Outlook 4

Building Resilient, Investable, and Tokenized Supply Chains 4

Case in Point: DBS, SGTraDex, & the Next Step—Stablecoin Settlement 5

5. EMERGING COUNTRIES IMPACT 5

How stablecoins support stability and welfare 5

Link to sustainable investment and SDGs 6

6. CONDITIONS AND RISKS 6

7. CONCLUSION AND POLICY RECOMMENDATION: 7

Stablecoins as the Operational Driver, CBDCs as the Sovereign Anchor 7

8. ABOUT THE AUTHORS 10



1. Digital Trade Finance, Stablecoins, and Sustainable Investment

Promoting Transparency, Accountability, and Monetary Stability

Digital trade finance platforms enhance supply chain transparency by enabling real-time data exchange and traceable documentation, while stablecoins add a layer of monetary stability and settlement efficiency. Technologies such as blockchain and IoT make it possible to verify product origins, labor practices, and carbon footprints, and to link these data streams to stablecoin-based payments in a single, auditable environment.

Stablecoins, typically pegged to fiat currencies or asset baskets, provide a predictable unit of account for cross-border trade, reducing FX volatility risk for emerging market businesses that would otherwise face unstable local currencies and costly hedging. This stability supports ESG-aligned investors by lowering financial risk in sustainable supply chains and making cash flows and impact outcomes more reliable.

Reducing Carbon Footprints and Operational Waste

Investment strategies increasingly prioritize companies and platforms that reduce emissions and streamline energy use along supply chains. Digitalization of trade processes eliminates paper documentation, optimizes logistics, and lowers transaction-related emissions, while blockchain settlement rails remove layers of intermediaries and reconciliation steps that add operational complexity and energy overhead.

As stablecoin-powered cross-border payments can settle within minutes and often at significantly lower fees—frequently under 1%—they reduce the need for capital-intensive pre-funding and repeated FX conversion, indirectly lowering the resource intensity of global trade finance operations. This trend contributes directly to investors' environmental objectives and portfolio decarbonization strategies by aligning financial flows with digital efficiency and reduced waste.

Supporting Green Finance, Programmability, and Incentive Alignment

Digital platforms enable financial institutions to incorporate sustainability metrics into credit assessments and transaction conditions, and stablecoins add programmability to how value is moved and controlled. Smart contracts can link stablecoin disbursements, trade loans, or discounting to verified ESG data from IoT devices and certified documents, triggering preferential financing rates for compliant, sustainable businesses and projects.



Investors can thereby align capital allocation with sustainability-linked instruments such as green trade loans, ESG-tied tokenized bonds, or receivables that settle in stablecoins when predefined environmental or social performance thresholds are met. This programmable alignment reinforces a positive feedback loop between ESG adherence, capital accessibility, and efficient, low-friction settlement.

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2. Combating Greenwashing Through AI, Data Analytics, and Tokenized Flows

One of the most pressing risks for sustainable investors is greenwashing—the exaggeration or falsification of ESG credentials. AI-driven solutions mitigate this issue by applying automated due diligence processes that cross-check supply chain, environmental, and social data against claims embedded in trade documentation and on-chain transactions.

When combined with blockchain and stablecoins, AI can monitor tokenized trade flows end-to-end, flagging anomalies between physical shipments, IoT readings, and payment events, thereby reducing informational asymmetry and strengthening investor confidence in ESG disclosures. For investors, this enhances portfolio quality, improves risk management, and ties actual capital movements—denominated in stablecoins—to verifiable sustainability outcomes.

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3. Stablecoins, Food Security, and Investment Risk in Commodity Trade

Food commodity trade finance presents a critical intersection between sustainability, financial inclusion, and investment risk. Over 50% of smaller trade transactions in food commodities are rejected due to escalating compliance costs and rigid due diligence requirements, contributing to underfinancing that exacerbates food scarcity and regional instability.

In this context, stablecoins can function as the settlement layer that simplifies cross-border payments for smaller counterparties and cooperatives, reducing friction from correspondent banking and FX constraints that disproportionately affect smaller, high-impact actors in agricultural supply chains. By embedding automated checks, tokenized documents, and ESG-linked payment conditions, digital platforms can expand food trade liquidity while offering investors a combination of financial returns, hunger mitigation, and political risk containment.



4. Stablecoins, SMEs, and Emerging Markets: Inclusion & Sustainability

Lowering Barriers to Global Trade Participation

Many SMEs in emerging markets are excluded from traditional trade finance due to limited collateral, thin credit files, currency volatility, and high costs of cross-border transactions. Stablecoin wallets accessible via mobile devices allow smaller traders to receive and send digital value directly, often without needing to open foreign currency accounts or rely on costly intermediary banks.

This infrastructure lowers entry barriers by offering:

- Faster, near–real-time settlement of cross-border invoices with fewer intermediaries.
- Reduced FX risk via holding and transacting in a stable, widely recognized digital currency.
- Improved cash flow predictability and working capital management for small exporters and importers.

These features directly support SDG-aligned objectives around economic inclusion, decent work, and reduced inequalities.

Financial Inclusion, ESG, and Digital Trade Ecosystems

Emerging markets with high mobile penetration but low banking density are particularly well-positioned to benefit from stablecoin-enabled digital trade ecosystems. As SMEs integrate into blockchain-based marketplaces and trade finance platforms, stablecoins serve as a neutral digital base currency that connects local economies to global buyers, investors, and logistics providers, while preserving transparent, traceable transaction records.

Backing platforms that enable underserved small enterprises to access stablecoin-based trade finance creates measurable social impact—expanding employment, income diversification and inclusive entrepreneurship—while opening access to high-growth markets. In this way, inclusion itself becomes a sustainability pillar that is tightly coupled to digital finance infrastructure.



Fintech Collaboration, CBDCs, and the Sustainable Investment Outlook

The convergence of fintech innovation and traditional finance is redefining sustainability-driven investment strategy. Collaborative models now integrate stablecoins, CBDCs, blockchain, IoT, and AI into trade finance workflows, creating interoperable ecosystems where public and private digital monies coexist.

Several emerging markets are piloting CBDCs and exploring regulated stablecoins to overcome currency volatility and banking gaps, while advanced economies design regulatory frameworks that require robust reserves, transparent governance, and strict compliance for stablecoin issuers. For investors, platforms that successfully integrate these instruments into trade flows offer both lower settlement and credit risk and clearer regulatory pathways, strengthening the long-term investability of sustainable trade ecosystems.

Building Resilient, Investable, and Tokenized Supply Chains

Digital trade finance systems are vital for building resilient supply chains that withstand production and logistics disruptions. Enhanced data integrity across IoT devices, digital documents, and on-chain transactions allows investors to model supply chain risks more accurately and to identify high-performing sectors aligned with sustainability goals.

When combined with stablecoin settlement, these systems:

- Enable real-time monitoring of both physical flows and financial flows in a unified data environment.
- Reduce fraud, errors, and compliance delays by linking tokenized trade documents directly to smart contract-enabled settlements.
- Improve trust among global buyers, suppliers, and investors by ensuring that ESG-aligned practices and outcomes are reflected in traceable, stablecoin-denominated transactions.

This integration enhances long-term investment stability and creates new ESG-linked digital assets—such as tokenized invoices, inventory, or carbon credits—denominated and settled in stablecoins.



Case in Point: DBS, SGTraDex, & the Next Step—Stablecoin Settlement

A notable example of sustainability-aligned digital infrastructure is DBS Bank's live transaction on Singapore's Trade Data Exchange (SGTraDex), where digitalization of bunkering documentation enabled real-time transaction validation and auditable ESG compliance. Building on such infrastructures, the next evolutionary step is integrating regulated stablecoins or tokenized deposits as settlement media, allowing trade participants to benefit simultaneously from transparent data and efficient digital value transfer.

SGTraDex demonstrates that much of the required digital infrastructure already exists. The next logical step is the integration of regulated digital money, allowing data, documentation and settlement to become part of a single programmable trade ecosystem.

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5. Emerging Countries Impact

Can Stablecoins Drive Sustainable Investment, Economic Stability and Welfare in Emerging Economies?

Yes—under the right regulatory and design conditions, stablecoins can be a significant driver for sustainable investment, economic stability, and welfare in emerging economies, but they are not a standalone solution and carry non-trivial risks.

How stablecoins support stability and welfare

Stablecoins can help households and smaller traders in high-inflation, currency-volatile environments preserve value and transact in a predictable unit of account, especially where access to foreign currency and formal banking is limited. In some African markets, for example, USD-pegged stablecoins already account for a large share of crypto transaction volume and are used by small businesses to pay suppliers and hedge against devaluation.

By lowering remittance costs and speeding cross-border transfers, stablecoins can increase the net income that migrant families and communities receive, which supports consumption, education, health, and small business investment. Research on blockchain-based remittances suggests that fee reductions from traditional levels above 6–8% to roughly 1–3% can materially enhance poverty reduction and local development outcomes.



LINK TO SUSTAINABLE INVESTMENT AND SDGS

Global development and policy institutions see stablecoins and related digital currency technologies as potential building blocks for a more sustainable, inclusive, and resilient financial system when properly governed. Lowering payment frictions and FX risk can crowd more capital—both remittances and formal investment—into underserved markets, advancing Sustainable Development Goals related to financial inclusion, decent work, and reduced inequalities.

For impact and ESG investors, stablecoin rails can make small-ticket, cross-border investments and shorter tenor trade finance more viable by reducing transaction costs and enabling better tracking of how funds are used. In principle, this can channel more private capital into green projects, smallholder agriculture, and inclusive digital businesses in emerging markets.

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6. Conditions and risks

Sustainability is no longer simply a compliance obligation; it has become a core investment criterion. Digital trade finance accelerates this transition by embedding ESG principles directly into trade flows. Within this framework, regulated stablecoins can provide a trusted settlement layer that expands access to finance for SMEs while improving transparency and operational efficiency.

For investors, such innovations offer transparency, efficiency, and a clear pathway to allocate capital responsibly toward businesses and platforms that substantively contribute to sustainable development, financial inclusion, and resilient supply chains. As trade digitalization accelerates, sustainable investment strategies will increasingly rely on the interplay of stablecoins, CBDCs, AI, and IoT to harmonize digital efficiency with environmental and social purpose.

At the same time, central banks and standard-setting bodies highlight that large, poorly regulated stablecoin ecosystems can create new financial stability risks, including runs on issuers and pressures on local monetary sovereignty. Without robust reserve management, consumer protection, AML/CFT controls, and interoperability with domestic payment systems, stablecoins can exacerbate, not reduce, vulnerabilities.



For stablecoins to genuinely act as a driver of sustainable investment and welfare in emerging countries, they need:

- Clear, enforceable regulation and supervision of issuers.
- Strong reserve transparency and risk management.
- Integration with domestic financial inclusion and digital ID strategies.
- Alignment with broader reforms in credit markets, trade finance, and social protection.

So, stablecoins can be a powerful enabler and accelerator of sustainable, inclusive growth, but only as part of a broader, well-governed policy and investment framework rather than as a standalone fix.

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7. Conclusion and Policy Recommendation:

Stablecoins as the Operational Driver, CBDCs as the Sovereign Anchor

The transformation of global trade finance is no longer theoretical—it is underway. Digitalization, ESG integration, tokenization of assets, and programmable financial flows are redefining how capital moves across borders. Within this transformation, regulated stablecoins have emerged as the most immediately deployable and scalable instrument to reduce settlement friction, mitigate FX volatility, and extend trade finance access to underserved businesses in emerging markets.

We therefore recommend that regulated, fully reserved stablecoins be recognized as the primary operational settlement layer for digital trade finance ecosystems, particularly in emerging markets where correspondent banking inefficiencies, currency volatility, and SME exclusion remain structural barriers to growth.

Regulated stablecoins improve trade finance by:

- Reducing settlement friction
- Near–real-time cross-border settlement
- Releasing trapped liquidity
- Lowering idle capital requirements
- Reducing foreign exchange uncertainty
- Predictable unit-of-account stability for SMEs
- Lower transaction and hedging costs



Together, these improvements enhance capital efficiency across global trade, enabling capital to reach more businesses, circulate more quickly, and support broader economic participation.

When embedded within digital trade platforms, stablecoins materially improve working capital efficiency, unlock smaller-ticket trade flows, and crowd private capital into sustainable supply chains. For ESG and impact investors, they provide the missing monetary layer that aligns capital efficiency with measurable sustainability outcomes.

However, sustainable financial architecture cannot rely solely on privately issued digital money.

Central Bank Digital Currencies (CBDCs) should serve as the sovereign-stable alternative and regulatory anchor within this ecosystem. While stablecoins provide agility, innovation, and market responsiveness, CBDCs provide monetary authority, systemic stability, and policy coherence. A well-designed framework should enable interoperability between regulated stablecoins and wholesale CBDCs, ensuring:

- Monetary sovereignty is preserved
- Systemic risk is contained
- Settlement finality is backed by central bank infrastructure
- Domestic financial inclusion strategies remain aligned with national development goals

In this hybrid model:

- Stablecoins act as the scalable, programmable “working layer” of global trade finance.
- CBDCs act as the stability backstop and sovereign settlement benchmark.
- Regulatory oversight ensures reserve transparency, interoperability, AML/CFT compliance, and investor protection.

For emerging economies, this balanced architecture offers the most credible pathway to enhance welfare, expand SME participation, stabilize trade flows, and attract long-term sustainable investment.

The recommendation is therefore not a binary choice between stablecoins and CBDCs. It is a structured coexistence:

Privately issued, well-regulated stablecoins to drive operational efficiency and inclusion — Publicly issued CBDCs to guarantee systemic stability and monetary trust.



Under robust governance, this dual-rail framework can materially narrow the global trade finance gap, strengthen food and commodity security, and accelerate progress toward the Sustainable Development Goals.

The future of trade finance will not be built on paper-based correspondent banking, but on digital money that is trusted, programmable and interoperable. Regulated stablecoins are best positioned to become the operational layer of that future, while CBDCs provide the sovereign foundation that ensures long-term monetary stability. Together they represent complementary pillars of a more resilient global trading system.



8. About the Authors

CHRIS SUNDERMAN

Chris is a member of the Board of Governors to TOTTA. He has a longstanding banking experience in client coverage, trade product and trade sales, covering commodity traders and wholesale clients. Chris is a trade innovator and the co-founder and co-owner of start-up companies who aim for an accessible and sustainable future for trade, finance & logistics for SMEs. Chris has the ambition to transform global trade by leveraging knowledge and experience of thought leaders in banking, trade finance, treasury and innovation to digit(al)ize for a sustainable and seamless trade ecosystem. Chris is a partner with T3i Partner Network, a global collective of industry-leading specialists in trade, treasury and technology with a focus on digital innovation.

SUBRA SHANKHAR

Subra is the founder and Principal of TOTTA, a tech industry veteran with a track record of 35 years successfully building and running businesses across the entire Asia Pacific. He has grown in the past decade as a respected voice in digitisation of trade. He is also a firm believer the role of non-banking financial institutions in Trade Finance will continue to grow significantly by 2030 for us to fill the gap. In this regard he is a Senior Advisor to TradeFlow Capital Management Pte Ltd. Subra, in addition, is a partner with T3i Partner Network, a global collective of industry-leading specialists in trade, treasury and technology with a focus on digital innovation.

TOTTA

TOTTA is an alliance of independent thought leaders in Global Trade from around the world. Its purpose is to increase global trade and reduce the gap in trade finance, towards UN's SDGs through accelerated adoption of technology. As an independent think tank TOTTA produces objective documents such as discussion papers & white papers, holding forums for robust exchange of ideas and publishing open-source software libraries.

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